Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-3296
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Team Size	4
Team Leader	Kamalesh P R
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current ComicCraft solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Current version is mainly focused on single comic generation and does not include user account management.	Limits personalized storage and long-term access to generated comics.	High
2	AI-based story and image generation can require significant processing time and resources.	May affect response time when usage increases.	High
3	The current deployment is designed primarily for local use.	Large-scale public access and distributed deployment require additional infrastructure.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed mainly for local/small-scale usage.	Deploy the FastAPI application on scalable cloud infrastructure and use load balancing or multiple application instances for increased concurrent users.
2	Data Storage	Generated comics and PDF files are stored within the project environment.	Introduce cloud/object storage and a database for user profiles, comic metadata, generated content, and downloadable files.
3	Performance	AI generation is performed during the comic creation workflow.	Use asynchronous/background processing, caching, optimized model usage, and queue-based processing to reduce waiting time under heavy load.
4	Security	API keys are configured through environment variables.	Add authentication, authorization, secure API-key management, input validation, rate limiting, HTTPS, and monitoring for production deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User accounts and profiles; save and manage generated comics.	Short Term	Personalized experience and persistent access to user-created comics.
Phase 2	Cloud deployment, database integration, and scalable file storage.	Medium Term	Support for increased users, data, and reliable access.
Phase 3	Advanced comic customization, multi-page story arcs, and comic libraries.	Medium to Long Term	Richer storytelling and improved customization.
Phase 4	Performance optimization, monitoring, stronger security, and expanded AI/model integration.	Long Term	More efficient, secure, and extensible ComicCraft platform.